

SIPs as a building Methodology



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By New Forest SIPs



Our Workshop is nestled in the New Forest near the picturesque coastal town of Lymington. Our rural location at Lea Green Farm, where we have been for over 20 years, is a constant inspiration for our low environmental impact, sustainable building methods.

Here at New Forest SIPs we have a highly experienced and motivated team who, from design phase to installation, produce houses of unparalleled quality.

We have worked hard to ensure that we are able to provide the highest standard of SIPs product available in the UK. However, providing an outstanding product is only one half of the story. To capitalise on the thermal values and natural airtightness of SIPs it must also have an exceptional installation team behind it. This is where we set ourselves apart by having the best available product and an elite, in-house installation team.

SIPs as a building methodology

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- **Airtightness and thermal performance**

Our product is naturally airtight and provides an insulated building envelope which acts as a thermal block. Air loss is thermal loss. Our product offers better than building regulations u-values for slimmer total wall buildups. This allows Clients to achieve long term energy savings within their homes as well as energy sustainability.

The unified structure and insulation of our factory-controlled product also eliminates the need for on-site trades insulating additionally.

- **Building speed and accuracy**

Due to the prefabricated nature of a SIPs build, we offer a combination of speed and precision not achievable through traditional building methods.

Through our design and manufacturing process we can guarantee dimensional accuracy, allowing the preorder of glazing elements which benefits the overall project timeline.

- **Cost confidence**

We ensure our costs and scope of works are made clear to Clients. When given architectural drawings, we are able to provide accurate quotations for our SIPs package, ensuring there are no hidden or spiralling costs.

Our quotation service provides a range of specifications to give clients full flexibility and choice to suit any budget.

- **Eco Sustainability**

Timber based construction is considered the most sustainable building methodology with the lowest CO2 impact of any other traditional building method. The embodied carbon of sustainably sourced timber is up to 82% less environmentally impactful. SIPs construction utilises timber extremely efficiently, using anywhere up to 70% less than a traditional timber frame build further lowering the carbon footprint of the build.

The Polyurethane (PU) core is CFC and HCFC free, has a GWP (Global Warming Potential) of less than 5 and saves more than 100x the energy used in its production. The high density and durability of the PU core provides unbeatable thermal benefits over the lifetime of the building.

A SIPs package is carefully designed with material efficiency in mind, ensuring the manufacture results minimal wastage further reducing the environmental impact of the build.

SIP Misconceptions

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Many Clients come to us having heard misconceptions about SIPs as a system. We are able to provide evidence to combat these misconceptions and reassure them that SIPs offer only advantages over traditional building methods.

Misconception	Reality
It isn't a tried and tested material and isn't as strong as brick.	SIPs have been in use in the UK for over 20 years and globally for much longer. Our panel is fully tested with a BBA accreditation giving it a minimum expected lifespan of 60 years and ensures the highest levels of strength and durability.
SIPs doesn't perform well in wet site conditions.	When installed in-line with our BBA accreditation our product does not suffer from exposure to wet conditions. We wrap each build in a high-performance breathable membrane allowing moisture to leave the system but not enter.
Is PU damaging as a material? What about off gassing?	PU is a closed-cell material that is non-irritant and does not contain any chemicals which disrupt the metabolism of living things. During the manufacture process "off gassing" happens as it cures, once this process is complete in the factory it is entirely inert and has no effect on human health.
Delamination of the panel and core material.	Our foam is auto-adhesively bonded to the OSB facing during the curing process to provide a unified product and therefore does not risk the same delamination as glued products such as EPS (polystyrene) SIPs.
Not an easily mortgageable product.	SIPs is considered standard construction and our BBA certificate in conjunction with our Gold accreditation form the structural timber association ensures it is fully insurable and mortgageable.

SIP Applications

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SIPs are a versatile product and can be used for a wide range of applications.

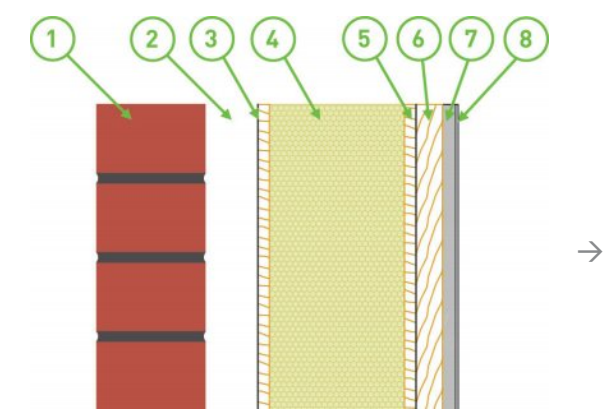
- New builds of all sizes - current building regulations to passive standard
- Extensions- single, two or three story - flat roof or pitched
- Garden rooms – offices, gyms and hobby rooms
- Annexes
- Class Q agricultural conversions (barn conversions)
- Mobile homes- caravan act applicable projects
- Roof replacements and roof lifts
- Commercial
- Large scale and repeatable projects
- Oak Frame overlaid



Project Gallery 001



SIPs Wall Build-ups



Number	Element	Width / mm
EXTERNAL		
1	External Finish*	20-102.5
2	Batten or air void	25-50l
3	Breather membrane	n/a
FABRIC		
4	SIP Wall	150-225
INTERNAL		
5	Foil VCL	n/a
6	Batten	25
7	Plasterboard	12.5
8	Plaster skim	2.5
TOTALS		235-417.5

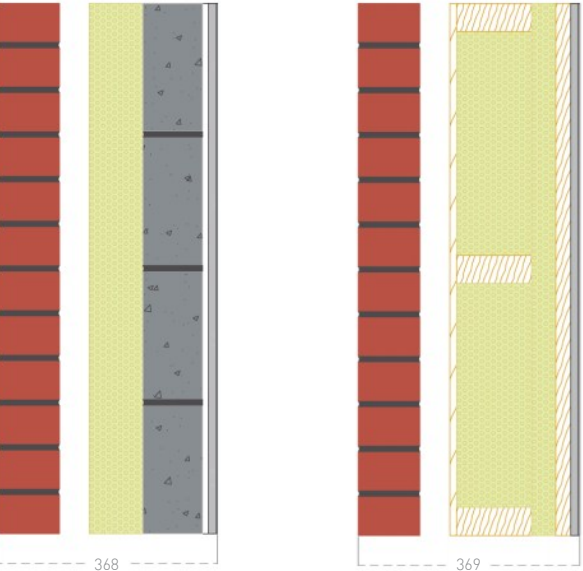
*External finishes include traditional full brick (102.5mm), brick slip (40mm+), render with carrier board (30-40mm) and timber cladding (20-30mm). All finishes are applicable.

As of the 15th of June 2022, the current building regulations detailed in Part L for new build domestic dwellings and extensions in England is 0.18W/m²K for walls.

TYPE	Wall Panel Thickness raw /mm	U-Value raw	Thickness finished* /mm	Finished u-Values
BASE	150	0.21	235-342.5	0.16
ADVANCED	175	0.17	260-367.5	0.14
ULTRA	200	0.15	285-392.5	0.12
ULTRA+	225	0.13	310-417.5	0.11

*External finishes include traditional full brick (102.5mm), brick slip (40mm+), render with carrier board (30-40mm) and timber cladding (20-30mm). All finishes are applicable.

Brick-faced Wall Build-ups



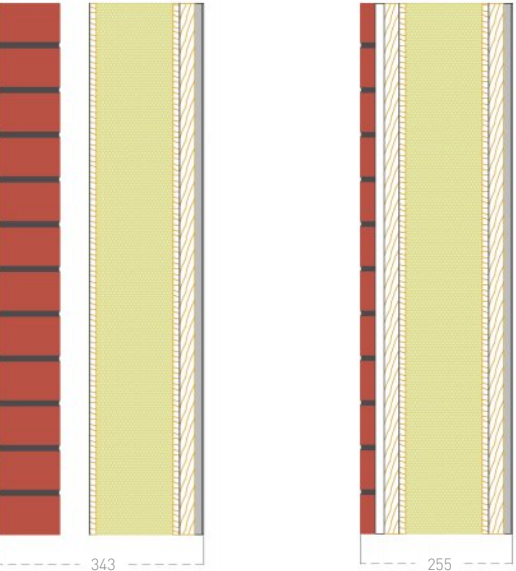
Traditional brick and block

Brickwork	102.5mm
Cavity	50mm
Insulation	90mm
Blockwork	100mm
Dot and dab	10mm
Plasterboard	12.5mm
Plaster skim	2.5mm
Total	367.5mm
u-Value	0.18W/m²K

Traditional brick and timber frame

Brickwork	102.5mm
Cavity	50mm
OSB	11mm
Stud work (insulated)	120mm
Insulation	40mm
Battens	25mm
Plasterboard	12.5mm
Plaster	2.5mm
Total	368.5mm
u-Value	0.18W/m²K

Brick + SIP Wall Build-ups



Brick faced SIP Wall

Brickwork	102.5mm
Cavity	50mm
SIPs*	150mm
Battens	25mm
Plasterboard	12.5mm
Plaster skim	2.5mm
Total	342.5mm
u-Value	0.16W/m²K

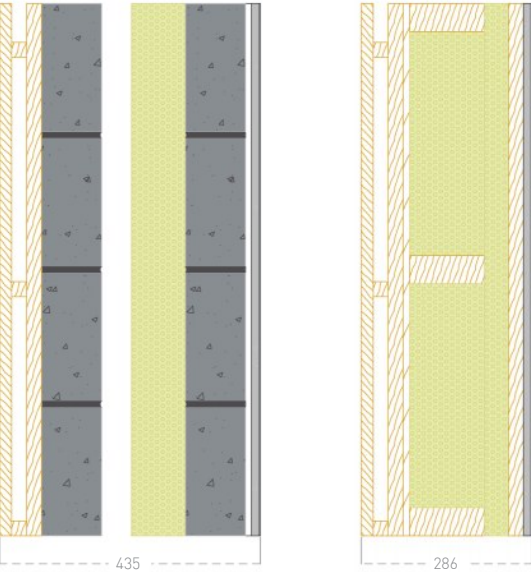
*GlideVale TF200 Thermo Membrane and VCL, Thickness negligible.

SIP wall with brick slip

Brick slip	20mm
Carrier board	15mm
Battens	25mm
SIPs*	150mm
Battens	25mm
Plasterboard	12.5mm
Plaster skim	2.5mm
Total	255mm
u-Value	0.16W/m²K

*GlideVale TF200 Thermo Membrane and VCL, Thickness negligible.

Timber Clad Wall Build-ups



Cladding on block

Cladding	20mm
Cross batten	50mm
Blockwork	100mm
Cavity	50mm
Insulation	90mm
Blockwork	100mm
Dot and dab	10mm
Plasterboard	12.5mm
Plaster skim	2.5mm
Total	435mm
u-Value	0.18W/m²K

Cladding on timber frame

Cladding	20mm
Cross batten	50mm
OSB	11mm
Stud work (insulated)	120mm
Insulation	40mm
Battens	25mm
Plasterboard	12.5mm
Plaster skim	2.5mm
Total	286mm
u-Value	0.18W/m²K

Cladding + SIP Wall Build-ups



→ Explained

Timber clad SIP wall offers slimmest build-up of 235mm total with u-Value 0.16

Compared to the Brick and block wall of 367.5mm, an increase of 56% over SIPs.

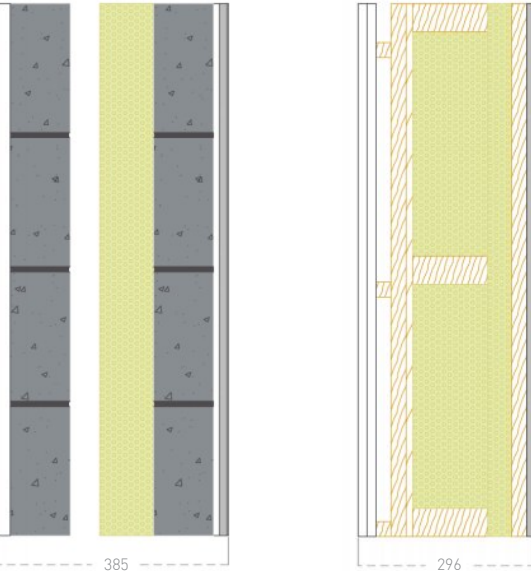
Slimmest timber frame wall is a timber clad timber frame wall, offering a 286mm build up. A 22% increase in thickness over SIPs.

Cladding on SIP wall

Cladding	20mm
Battens	25mm
SIPs*	150mm
Battens	25mm
Plasterboard	12.5mm
Plaster skim	2.5mm
Total	235mm
u-Value	0.16W/m²K

*GlideVale TF200 Thermo Membrane and VCL, Thickness negligible.

Render-faced Wall Build-ups



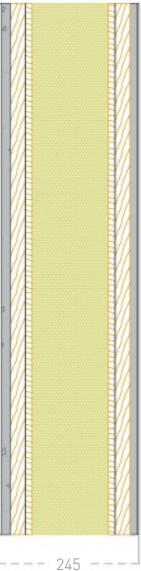
Render on block

Render	20mm
Blockwork	100mm
Cavity	50mm
Insulation	90mm
Dot and dab	10mm
Plasterboard	12.5mm
Plaster skim	2.5mm
Total	385mm
u-Value	0.18W/m²K

Render on timber frame

Render	15mm
Carrier board	15mm
Cross batten	50mm
OSB	11mm
Stud work (insulated)	120mm
Plasterboard	12.5mm
Plaster skim	2.5mm
Total	296mm
u-Value	0.18W/m²K

Render-faced + SIP Wall Build-ups



→ Explained

With a requirement u-Value of 0.18W/m²K for external walls on new builds, SIPs makes a considerable improvement to 0.16 at our 150mm panel thickness.

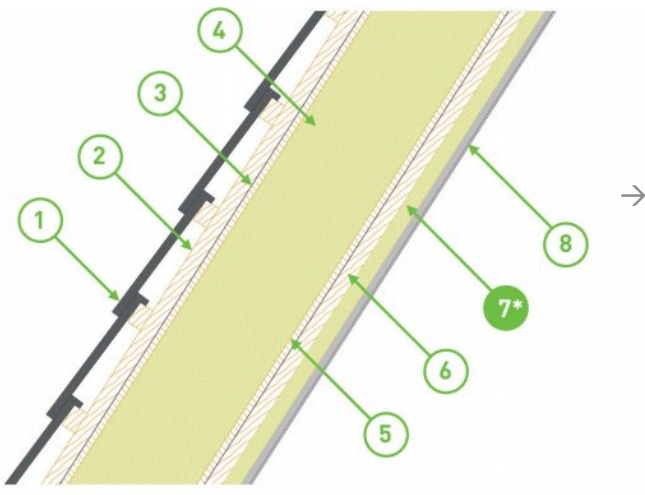
SIPs total build-up of 235 to 255mm for finished SIP wall around 1/3 slimmer compared to brick and block.

Render on SIP wall

Render	15mm
Carrier board	15mm
Battens	25mm
SIPs*	150mm
Battens	25mm
Plasterboard	12.5mm
Plaster skim	2.5mm
Total	245mm
u-Value	0.16W/m²K

*GlideVale TF200 Thermo Membrane and VCL, Thickness negligible.

SIPs Roof Build-up



Number	Element	Width / mm
EXTERNAL		
1	External Finish**	20+
2	Batten or air void	50
3	Breather membrane	n/a
FABRIC		
4	SIP Wall	175-225
INTERNAL		
5	Foil VCL	n/a
6	Batten	25
7	Plasterboard (insulated)	12.5 [37.5]
8	Plaster skim	2.5
TOTALS		285-360+

*Insulated plasterboard only required for new build projects; extensions do not necessarily need it.

**External finishes from slate, concrete tile, zinc, metal clad as well as integrated or bracket mounted solar panels.

New Build Options-Roof

As of the 15th of June 2022, the current building regulations detailed in Part L for new build domestic dwellings in England is 0.11W/m²K for roofs.

TYPE	Wall Panel Thickness raw /mm	U-Value raw	Thickness finished* /mm	Finished u-Values
BASE	200	0.15	335+	0.11
ADVANCED				
ULTRA	225	0.13	360+	0.10
ULTRA+				

*External finishes from slate, concrete tile, zinc, metal clad as well as integrated or bracket mounted solar panels.

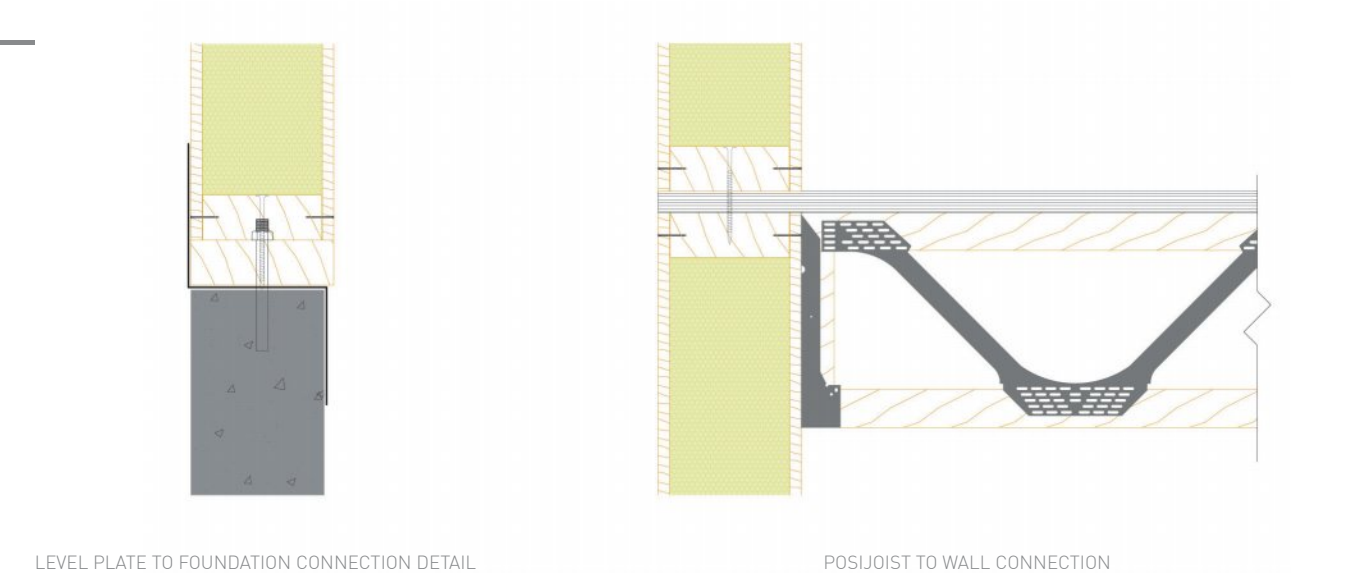
Extension Options-Roof

As of the 15th of June 2022, the current building regulations detailed in Part L for extensions to domestic dwellings in England is 0.15W/m²K for roofs.

TYPE	Wall Panel Thickness raw /mm	U-Value raw	Thickness finished* /mm	Finished u-Values
BASE	175	0.17	285+	0.15
ADVANCED	200	0.15	310+	0.13 - 0.11
ULTRA	225	0.13	335+	0.12 - 0.10
ULTRA+				

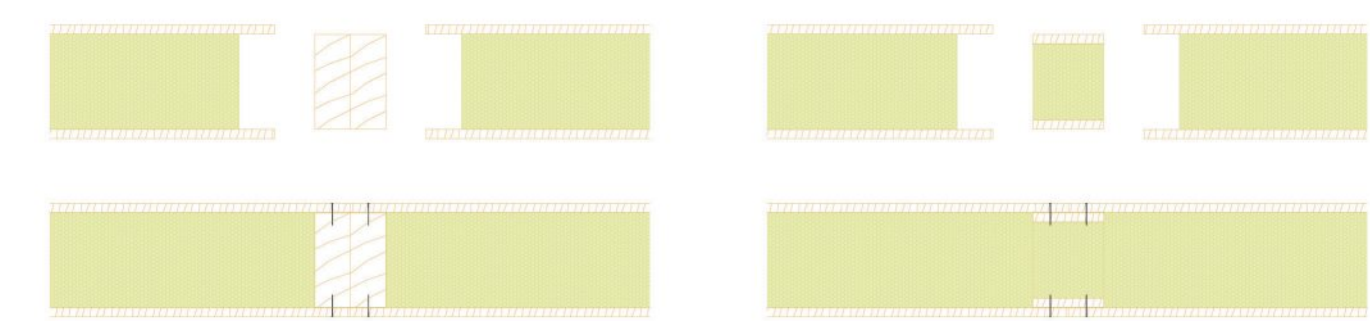
*External finishes from slate, concrete tile, zinc, metal clad as well as integrated or bracket mounted solar panels. Insulated plasterboard only required for new build projects; extensions do not necessarily need it.

CAD Drawing detail examples



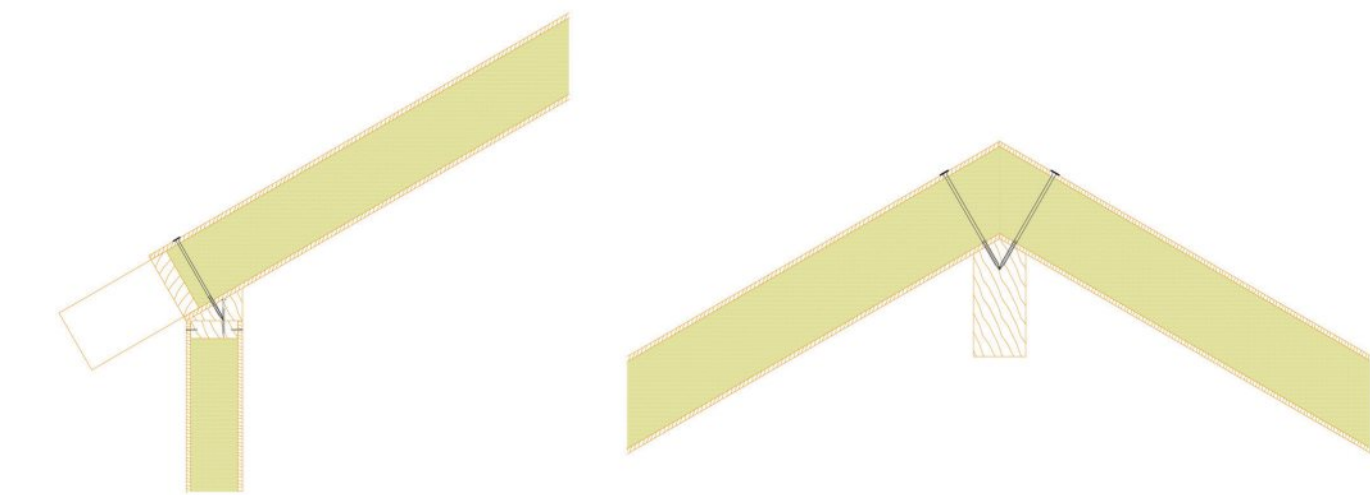
LEVEL PLATE TO FOUNDATION CONNECTION DETAIL

POSIJOIST TO WALL CONNECTION



TIMBER CONNECTION SPLINE DETAIL

SIP INSULATED CONNECTION SPLINE DETAIL



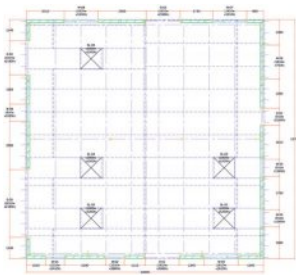
WALL TO PITCHED ROOF SECTION

GLULAM RIDGE DETAIL

Typical SIP Project Pathway

01 Design

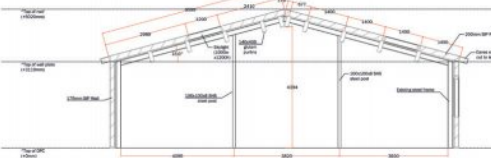
- Following order confirmation...
- Structural engineer instructed, completed in 1-4 weeks
 - SIP Sign Off Pack issued within 1-4 weeks
 - Sign Off Pack reviewed and signed off by the client or architect
 - Timescale between 2-9 weeks,



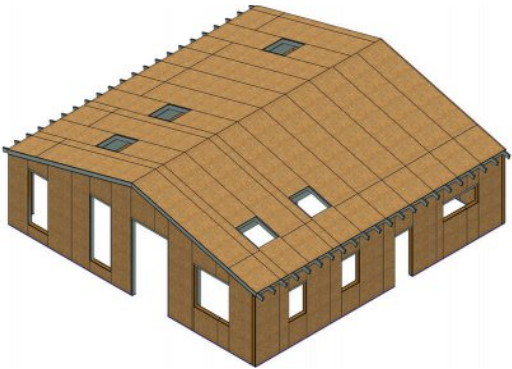
Plan drawings with dimensions including aperture sizes/positions



Relevant sectional drawings and relevant details



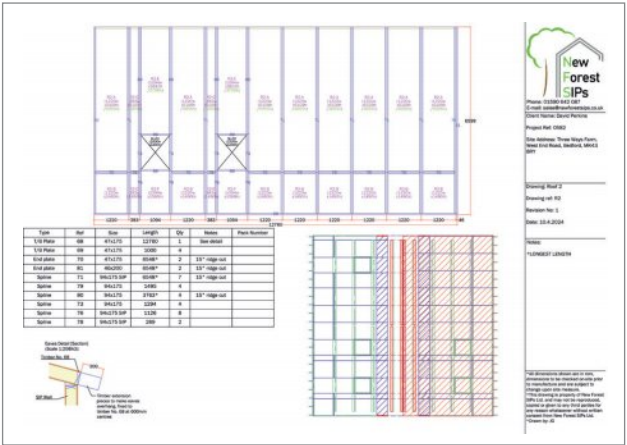
Elevations with aperture sizes and cill heights



Isometric views of the 3D model

02 Procurement

- Manufacture and installation drawings 2-4 weeks
- Material procurement 2-6 weeks (steelwork and joists included)



03 Manufacture

- Manufacture in our factory 2-4 weeks
- Delivery to site 1 week



04 Installation

- Installation
- Handover and completion
- SIPs structure ready for follow on trades to start



Build Stages



Stage 1: Plate down



Stage 3: Nail up



Stage 5: Final nail up



Stage 2: Walls & Structures



Stage 4: Roof on



Stage 6: Membrane

Costings per m²

— In the UK the average finished home (from dirt to completion) costs vary from £1,900+ per m² with the Client having a “hands-on” self building approach up to £3,200+ per m² for a more turnkey package.

SIPs compliments any of the chosen build routes, be it self-build, self-managed, main contractor or turnkey by providing the Client or contractor with decisive, fixed costings for the main superstructure of the building. On average SIPs pricing ranges from £300-£600 per m² with installation and structures included within this figure. The top end pricing tends to be attributed to projects with complex roofs, large overhangs or cantilevers, huge open-plan spaces and wide apertures which would be no different to other building methodologies.

The only varying cost is the steelwork as this is reliant on the structural engineering calculations performed at the first stage of the build process. For the quotation stage we provide a steelwork budget to the Client as a cost approximation and more often than not land within budget.

We offer different panel thicknesses with different price points to suit every kind of project. Our “Base” specification offers our most affordable package with all of the benefits of using the SIPs system. The thicker panels in the “Advanced”, “Ultra” and “Ultra+” specifications allow Clients to explore the possibility of moving a project toward Passivhaus standard. All these options are detailed within our comprehensive quotation document which aids in the decision making process.

Accreditations

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Case Study 001 / Project No 0462

Class Q Barn Conversion / Wimborne

— This 275m² barn conversion near Wimborne, Dorset is one of our favourite projects to date. With interesting vaulted elements, the integration with the existing concrete frame (which was part of the Class Q conditions) and open plan living areas it created a very calm building which is complimentary to its rural setting. The Clients took full advantage of the incredible views with some cleverly placed glazing throughout.



Case Study 002 / Project No 0258

New Build / Harpenden

— Designed to imitate the looks of a chicken barn which was once on the property, this new build achieved incredible results. We really enjoyed utilising our joinery routes and manufacturing some stunning oak trusses which sat in the open plan kitchen living area as a fantastic, traditional structural feature to this otherwise modern-styled build. The Client was aiming for an air tightness test result of 3, the build achieved 0.6!



Case Study 003 / Project No 229

Bungalow Roof Lift / Milford-on-Sea

— This 1920's bungalow renovation involved the addition of a new first floor and roof in SIPs. This doubled the liveable floor area of the property and provided a much needed thermal boost to an old, uninsulated building. The age of the existing building bought its own challenges, but our experienced fitting team completed the project in under 3 weeks. Where the SIP system is so light it meant that minimal repair works were required to the existing foundations.



Project Gallery 002



Thank you



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